

Alex Brookes

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TS/SCI Cleared | M.S. Computer Science (AI) | B.S. Aerospace Engineering

SUMMARY

Engineer with 6+ years of experience applying quantitative analysis, statistical modeling, machine learning, and predictive analytics to complex technical problems. M.S. in Computer Science (Artificial Intelligence) and B.S. in Aerospace Engineering with minors in Physics and Computer Science. Experience conducting research, developing predictive models, evaluating signals, and analyzing large-scale datasets under uncertainty.

EXPERIENCE

Aerospace Engineer | National Air and Space Intelligence Center (NASIC) | 2020–Present

- Applied quantitative analysis and statistical reasoning to evaluate complex aerospace systems, producing timely, actionable intelligence assessments supporting national security objectives.
- Investigated technical problems involving incomplete, uncertain, and high-volume data, synthesizing findings into technical products for operational decision-makers.
- Developed Python and C++ software tools to automate quantitative analysis, data processing, and technical evaluation workflows.
- Collaborated across multidisciplinary teams to translate ambiguous, real-world technical questions into structured analytical approaches.

EDUCATION

Air Force Institute of Technology (AFIT)

M.S. Computer Science (Artificial Intelligence), June 2026

GPA: 3.78/4.00

Relevant Coursework: Statistical Learning, Probability, Optimization, Algorithms, Deep Learning

University of Virginia

B.S. Aerospace Engineering, 2020

Minors: Computer Science, Physics

RESEARCH PROJECTS

ADS-B Bit-Level Denoising

- Designed and evaluated convolutional autoencoder architectures for reconstruction of aircraft position signals corrupted with 10–40% bit error rates, achieving ~35% and

~17% RMSE reduction over a Random Forest baseline at 10% and 20% BER, respectively.

- Investigated direct learning on IEEE-754 bit representations rather than conventional floating-point features, identifying exponent-bit recovery as a primary driver of improved performance.
- Conducted model evaluation across multiple noise regimes using a frozen evaluation manifest to ensure fair benchmarking against baseline approaches.
- Utilized TensorFlow, Keras, NumPy, Pandas, feature engineering, model validation, and statistical performance analysis.

Bitcoin Limit Order Book Modeling

- Implemented DeepLOB-style neural architectures on high-frequency BTC/USDT order book data, achieving a macro F1 of 0.532 on held-out test data versus a LightGBM baseline.
- Engineered features from market microstructure data and performed walk-forward validation to guard against lookahead bias.
- Compared deep learning models against gradient-boosted tree baselines using out-of-sample testing procedures.

Numerai Regime Ensemble

- Developed a regime-aware ensemble framework using K-Means clustering on per-era predictive signal vectors to identify distinct market regimes.
- Routed predictions to specialized regime-conditional models, achieving a +0.006 IC improvement over baseline on purged cross-validation.
- Applied unsupervised learning, ensemble learning, model selection, and cross-validation techniques.
- Identified a distribution-shift failure mode that impacted out-of-sample regime routing.

TECHNICAL SKILLS

Programming: Python, C++, Bash

Data Analysis: NumPy, Pandas

Machine Learning: TensorFlow, Keras, PyTorch, LightGBM, Deep Learning, Neural Networks, Autoencoders, Ensemble Learning, Feature Engineering

Statistics: Statistical Modeling, Predictive Modeling, Time Series Analysis, Hypothesis

Testing, Cross Validation, Walk-Forward Validation, Signal Processing, Market

Microstructure

Research: Experimental Design, Model Evaluation, Performance Benchmarking

Systems: Linux, Git, Docker, High Performance Computing

AWARDS

Category II Civilian of the Year (2023), Squadron and Group Winner; Wing Runner-Up

Active TS/SCI Clearance

Vice Chair, International Technical Conference